GCSE PHYSICS · TEACHING DECK

Energy Stores & Changes

Deck 1 of 4 — Systems, transfers & how to describe them

AQA 8463 | Edexcel 1PH0 | OCR J249 / J250

The bedrock of the whole topic — every energy question starts with naming a store correctly



Where we're going — 4 decks

YOU ARE HERE

1

Systems, transfers & how to describe them

All tiers

2

Kinetic & gravitational potential energy

All tiers

3

Elastic potential energy & bringing it together

Higher stretch

4

Quick recap & multi-step challenges

Review

A rare split-free topic — Part 1 is common to every tier, board and course (Combined and Separate). There is almost nothing to withhold from anyone here.

STARTER *~1 min*



"Clap your hands. You just made a sound — but energy is never made from nothing. Trace it back: where was that energy a moment before?"

Push toward: sound ← movement ← muscles ← food.

Tutor: *This is the whole deck's big idea — energy is never created, only stored, and it moves between stores.*

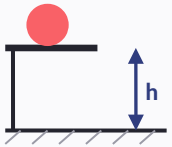
THEIR IDEAS

Systems & the eight energy stores

- A system is an object or group of objects you choose to study — e.g. a kettle and the water in it, or a ball and the Earth.
- Energy is never created or destroyed. It is stored, and it moves (transfers) between stores when a system changes.
- **The eight energy stores (learn all eight): kinetic, gravitational potential, elastic potential, thermal / internal, chemical, nuclear, electrostatic, magnetic.**
- Describe energy by where it is stored — not “light energy” or “sound energy”. Those are ways energy is transferred, not stores.



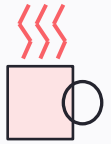
Kinetic
a moving car or ball



Gravitational potential
a book on a high shelf



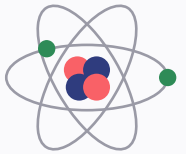
Elastic potential
a stretched spring or band



Thermal / internal
a hot cup of tea



Chemical
food, fuel, a battery



Nuclear
inside atomic nuclei



Electrostatic
separated + and – charges



Magnetic
magnets and iron

Where marks are lost: vague names (“heat energy” not “thermal store”); confusing a store with a transfer; forgetting chemical, nuclear, electrostatic and magnetic.

WORKED EXAMPLE

Tutor models the method — answer comes last

A stretched catapult holds a stone. Name the main store now, and just after release as the stone flies up.

METHOD

- The elastic potential store is full while the catapult is stretched.
- On release it empties into the kinetic store.
- As the stone rises, energy shifts to the gravitational potential store.

WORK IT THROUGH

Answer: Elastic potential (stretched) → kinetic (on release) → gravitational potential (as it rises).

Mark scheme: *1 for elastic potential · 1 for kinetic · 1 for gravitational potential.*

YOUR TURN — PART 1

Systems & the eight energy stores — answers on the next slide.

1 Define “system” in one sentence.

2 Name the store filling up: a car speeding up; a book lifted onto a shelf; a stretched elastic band.

3 For a cooling cup of tea, which store empties?

4 Why is “sound energy” not one of the eight stores?

Working space is squared on purpose — students calculate, tutor annotates.

ANSWERS — PART 1

Tutor reveal

1

A system is an object or group of objects being studied.

2

Kinetic · gravitational potential · elastic potential.

3

The tea's thermal store empties.

4

Sound is a way energy is transferred (by radiation), not a store where energy sits.

STARTER *~1 min*



"A ball bounces, but each bounce is a little lower than the last. Where is the energy at the top of a bounce, at the bottom — and where does the missing energy go?"

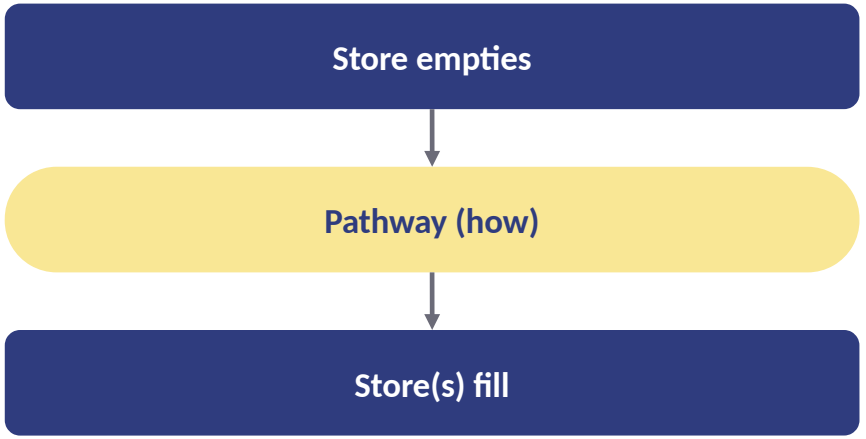
Push toward: GPE ⇄ KE each bounce, with a little energy dissipated (heating + sound) every time it hits the ground.

Tutor: *Sets up the "describe the energy changes" question type — core marks on every board.*

THEIR IDEAS

Four pathways, five situations to describe

- Energy transfers between stores by four pathways: mechanically (a force does work), electrically (a moving charge does work), by heating (a temperature difference), and by radiation (e.g. light, sound, infrared).
- **To answer "describe the energy changes": name the store that empties → the pathway → the store(s) that fill.**
- Five situations to know: projected upwards; hits an obstacle; accelerated by a force; vehicle braking; kettle boiling.



The shape of every "describe the energy changes" answer.

THE FOUR PATHWAYS

Mechanically
a force does work

Electrically
a moving charge does work

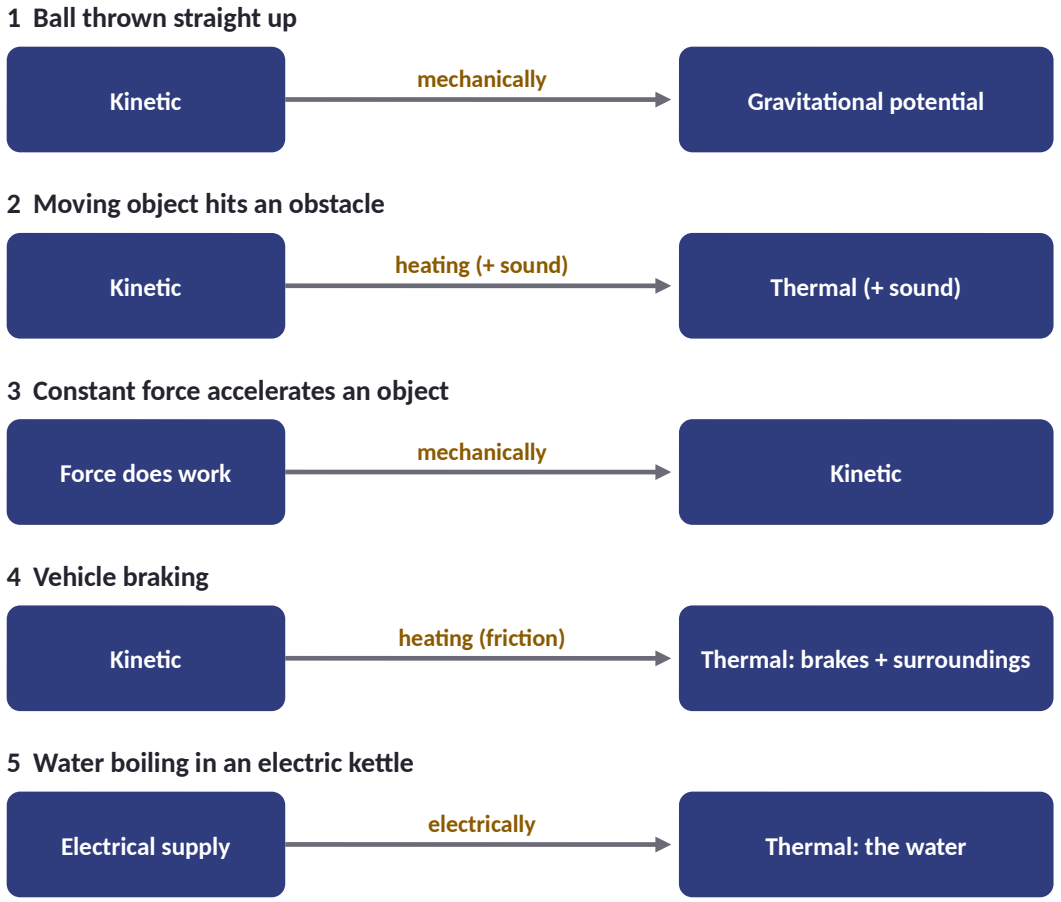
By heating
a temperature difference

By radiation
light, sound, infrared

Where marks are lost: "kinetic pathway" or "heat store" (pathway = how, store = where); saying energy is "used up" or "lost" instead of transferred.

The five situations, store by store

- Always write the chain in order: store that empties → pathway → store that fills.
- Where the “extra” energy goes matters: in 2 and 4 it is dissipated — transferred by heating to the thermal store of the surroundings (and some to sound). It is never “used up”.
- Use exact names: “thermal store”, “kinetic store”, “gravitational potential store” — not “heat”, “movement” or “height energy”.



Where marks are lost: naming only the store that fills and forgetting the one that empties (or vice versa).

WORKED EXAMPLE

Tutor models the method — answer comes last

A car brakes to a stop. Describe the energy transfer.

METHOD

- The kinetic store of the car empties.
- Energy is transferred by heating (friction in the brakes) to the thermal store of the brakes and the surroundings.

WORK IT THROUGH

Answer: Kinetic store empties → transferred by heating (friction) → thermal store of the brakes and surroundings fills.

Mark scheme: *1 for kinetic store empties · 1 for by heating (friction) · 1 for thermal store of the brakes / surroundings.*

YOUR TURN — PART 2

Transfer pathways & describing the changes — answers on the next slide.

1 State the four transfer pathways.

2 Describe the main energy changes when a ball is thrown straight up (ignore air resistance).

3 Describe the energy changes when a kettle boils water.

4 A wind-up toy runs across the floor. Name the store that empties and the pathway.

Working space is squared on purpose — students calculate, tutor annotates.

ANSWERS — PART 2

Tutor reveal

1

Mechanically, electrically, by heating, by radiation.

2

Kinetic store empties → gravitational potential store fills (then reverses on the way down).

3

Energy supplied electrically → thermal store of the water fills.

4

Elastic potential store of the wound spring empties → transferred mechanically to the kinetic store.

EXAM PRACTICE — SET A

Energy stores and transfers · 12 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q1 *Stores & transfers · table* **[6 marks]**

The table shows the energy stores involved in three situations.

Situation	Store that empties	Store that fills
A stretched catapult fires a stone	*elastic potential*	*kinetic*
(i) A student lifts a book onto a shelf		
(ii) A skateboard coasts to a stop		
(iii) A ball falls off a table		

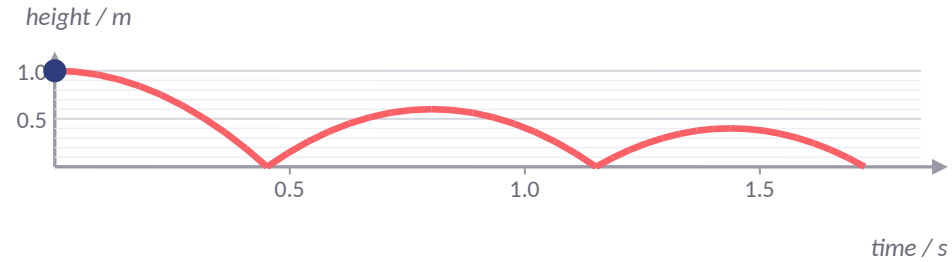
(a) Complete rows (i), (ii) and (iii) of the table. **[3]**

(b) Name the pathway that transfers energy to the surroundings in row (ii). **[1]**

(c) A student says: "The skateboard's energy has been used up." Explain why this is wrong. **[2]**

Q2 *Graph · describe & explain* **[6 marks]**

A student drops a ball. The graph shows its height as it bounces.



(a) Use the graph to find the height lost by the first peak. **[1]**

(b) Name the energy store that is greatest at each peak. **[1]**

(c) Describe the energy changes from release to the first impact. **[2]**

(d) Explain why each bounce is lower than the one before. **[2]**

MARK SCHEME — SET A

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q1

[6 marks]

B1 (i) chemical (store of the student) → gravitational potential

B1 (ii) kinetic → thermal

B1 (iii) gravitational potential → kinetic

Each row needs BOTH stores correct. Accept "internal" for thermal. "Heat energy", "movement energy": no credit.

B1 (b) by heating (accept "heating by friction")

B1 (c) energy is not used up / destroyed — the total stays the same (conserved)

B1 it is transferred (dissipated) by heating to the thermal store of the surroundings (wheels, road, air)

Do not accept "lost" or "wasted" on their own.

Answer: (a) chemical → gravitational potential; kinetic → thermal; gravitational potential → kinetic

Q2

[6 marks]

B1 (a) 1.0 − 0.6 = 0.4 m (allow 0.38–0.42)

B1 (b) gravitational potential (store)

B1 (c) gravitational potential store empties

B1 kinetic store fills (transferred mechanically, by the force of gravity)

B1 (d) on each impact some energy is dissipated / transferred by heating (and sound) to the thermal store of the ball, floor and surroundings

B1 so less energy is left in the kinetic store after the bounce, so less is transferred to the gravitational potential store at the top

Do not accept "energy is lost" or "used up" without saying where it goes.

Answer: (a) 0.4 m (b) gravitational potential

EXAM PRACTICE — 6-MARK QUESTION

Levels-based · 6 marks in total · marks shown in [brackets] · mark scheme on the next slide

Q3 *Extended response · levels* **[6 marks]**

*A child winds up a clockwork toy car, places it on the floor and lets go. The car speeds across the floor and gradually slows to a stop. Describe the energy transfers that take place, from the child winding up the toy until the car stops. Name each store and each pathway you use.

MARK SCHEME — 6-MARK QUESTION

Tutor reveal · M = method, A = accuracy, B = independent mark, ECF = error carried forward

Q3

[6 marks]

Level 3 (5–6) a complete, linked sequence covering winding, release AND slowing, naming stores and pathways correctly, with the energy ending in the thermal store of the surroundings.

Level 2 (3–4) most of the sequence, or the whole sequence with a store / pathway missing or misnamed (e.g. “heat”).

Level 1 (1–2) isolated correct stores or transfers, no clear sequence.

Indicative content: chemical store of the child’s muscles empties; energy transferred mechanically (work done winding) to the elastic potential store of the spring; on release the elastic potential store empties, transferred mechanically to the kinetic store of the car; friction between wheels / floor and air resistance: kinetic store empties, energy transferred by heating to the thermal store of the wheels, floor and air (dissipated); some energy transferred by sound (radiation); total energy conserved — none used up.

Max 6 marks

Answer: **Levels-based: see descriptors**

Watch out — and what to remember

IN A NUTSHELL

- Energy is stored, never created or destroyed.
- Learn all eight stores — describe by where it's stored, not by the transfer.
- Every “describe the changes” answer: store empties → pathway → store fills.

COMMON SLIP-UPS

Calling a transfer a store

“Sound energy” and “light energy” are pathways (radiation), not stores. Only the eight named stores count.

Forgetting the less obvious stores

Chemical, nuclear, electrostatic and magnetic are the ones students miss under pressure — drill all eight, not just kinetic and GPE.

Saying energy is “used up”

Energy is never lost or used up — always name where it goes and by which pathway.

NEXT Deck 2 — Kinetic & gravitational potential energy